

Impact of SMOTE-Based Class Rebalancing on SHAP and LIME Explanation Quality in Random Forest Network Intrusion Detection

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Abstract—Machine learning-based Network Intrusion Detection Systems (NIDS) routinely apply the Synthetic Minority Oversampling Technique (SMOTE) to address severe traffic-class imbalance, yet the impact of such rebalancing on post-hoc explainability outputs has not been systematically quantified. This study investigates how SMOTE rebalancing affects feature attribution rankings produced by SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) in a Random Forest NIDS trained and evaluated on the UNSW-NB15 benchmark. Two classifiers — one trained on the original imbalanced partition and one on a SMOTE-rebalanced partition — are evaluated against the same 82,332-instance test set using a statistical framework comprising McNemar’s test, Wilcoxon signed-rank tests, bootstrap confidence intervals, and Holm–Bonferroni correction. SMOTE improves macro-averaged F1 and substantially increases minority-class recall, with all aggregate predictive effect sizes remaining negligible (Cohen’s $h < 0.09$). In contrast, SHAP attribution rankings shift with a medium effect ($r = 0.33$) and LIME rankings with a large effect ($r = 0.60$), demonstrating that class rebalancing alters explanation outputs disproportionately relative to classification metrics. Accuracy-only validation after NIDS retraining is insufficient to characterise post-hoc explanation stability.

Index Terms—network intrusion detection, explainability, class imbalance, synthetic minority oversampling, Shapley additive explanations, local interpretable model-agnostic explanations, random forest, feature attribution

I. INTRODUCTION

Machine learning-based NIDS have demonstrated strong performance on curated benchmarks [1]–[3], yet a persistent challenge is severe class imbalance: benign traffic dominates by orders of magnitude while rare attack categories may constitute fewer than 0.1% of packets [4], [5]. This asymmetry biases classifiers toward majority-class predictions and suppresses detection of rare, high-severity attacks.

SMOTE [6] is the dominant remediation strategy, generating synthetic minority-class instances by interpolating between existing samples in feature space. SMOTE has been shown to improve minority-class recall in NIDS applications [1], [7]; in this study it increased Worms-class F1 from 0.233 to 0.452.

Concurrently, eXplainable AI (XAI) adoption in security operations has grown substantially [8], [9]. As AI systems

become subject to regulatory transparency requirements [23], the reliability of post-hoc explanations across model updates is critical. SHAP [10], [11] and LIME [12]–[14] are the two most widely applied post-hoc explanation methods for NIDS models.

A critical and underexplored question is: *when a NIDS model is retrained with a different class distribution, does the explanation change as much as the prediction?* If explanation instability is disproportionate to predictive instability, operators relying on accuracy alone to validate retrained models may unknowingly deploy systems whose explanation outputs have changed substantially, undermining the trustworthiness of XAI-assisted security analysis.

This paper addresses that gap with a controlled empirical study. The contributions are:

- 1) A paired statistical comparison (McNemar’s test, Wilcoxon signed-rank, 95% bootstrap CI) of Baseline and SMOTE-trained Random Forest models on UNSW-NB15 (82,332 test instances).
- 2) Quantified effect sizes for predictive metrics (Cohen’s h) and explanation stability (rank-biserial r), with explicit effect-size interpretability bounds.
- 3) A systematic comparison of SHAP and LIME sensitivity to training class distribution, demonstrating that LIME is substantially more sensitive.
- 4) Actionable guidance on explanation-consistency monitoring in XAI-augmented NIDS pipelines.

II. RELATED WORK

A. Class Imbalance in NIDS

Class imbalance is a well-documented challenge in NIDS [4], [5]. Minority attack categories are rare relative to normal traffic, causing classifiers to achieve high accuracy by predicting the majority class [1], [6]. Re-sampling strategies — including SMOTE, ADASYN, and random undersampling [5] — along with cost-sensitive learning and ensemble methods [5] have been proposed. SMOTE remains dominant due to its consistency and reproducibility [2], [7]. UNSW-NB15 [4] is

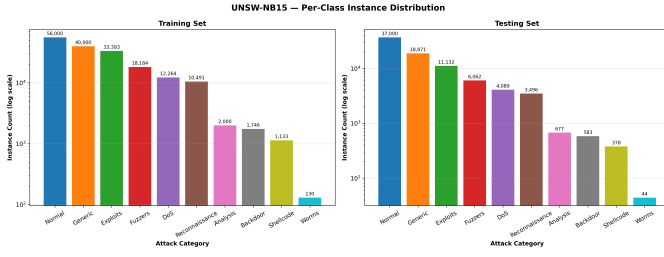


Fig. 1. Training set class distribution before SMOTE (175,341 instances, 10 classes, log scale). Maximum imbalance: 430.77:1 (Worms: 130 vs. Normal: 56,000). After SMOTE: 56,000 per class.

the established benchmark used here, selected for its predefined train/test split and role as a community evaluation standard [3], [14].

B. Explainable AI for Network Security

XAI application to NIDS has expanded alongside the trustworthy-AI movement [8], [9]. SHAP provides attributions with theoretical guarantees of consistency and local accuracy [10], [11]. LIME approximates model behaviour locally with an interpretable surrogate without requiring structural knowledge of the model [12]. Both have been applied to Random Forest [14], [15] and deep learning-based NIDS [13].

C. XAI Stability and Explanation Quality

Explanation stability under perturbation has been studied in general ML contexts [16] but remains underexplored in NIDS settings. Prior work examined adversarial perturbation effects on LIME and SHAP [8] and sensitivity to hyperparameters [16]. The specific question of how training data rebalancing affects explanation stability — distinct from prediction stability — has not been addressed in prior NIDS literature [17], [18]. This study addresses that gap directly.

III. METHODOLOGY

A. Dataset

The UNSW-NB15 dataset [4] provides a predefined training set (175,341 instances) and test set (82,332 instances) across 10 classes. The training set exhibits severe imbalance (Fig. 1): Normal contains 56,000 samples (31.9%) while Worms contains only 130 (0.074%), yielding a maximum imbalance ratio of 430.77:1. UNSW-NB15 was selected for its controlled-environment realism, predefined split, and status as a community benchmark for reproducible NIDS evaluation [3].

B. Feature Engineering

The original 44-feature schema was reduced to 42 features by removing `id` (identifier) and `label` (target leakage). No scaling was applied; Random Forest is scale-invariant.

C. Class Rebalancing

SMOTE [6] was applied with `sampling_strategy='auto'`, `k_neighbors=5`, seed 42. Minority classes were oversampled to 56,000 per class, producing 560,000 training rows (384,659 synthetic samples). SMOTE was selected as the most widely benchmarked NIDS rebalancing technique [1], [2], enabling direct comparison with prior work. The test set was not modified; both models are evaluated on the same held-out set.

D. Classifier

A Random Forest [19] — selected for strong NIDS benchmark performance and native compatibility with SHAP TreeExplainer — was trained independently on the Baseline (imbalanced) and SMOTE-balanced datasets. All hyperparameters and seed (42) were held constant. The original train/test split was used without re-splitting.

E. XAI Methods

SHAP: SHAP values [10], [11] were computed for 60 stratified test instances (six per class) using `shap.TreeExplainer`, yielding exact Shapley values. Global importance was obtained by averaging $|\text{SHAP}|$ values. Rankings were compared using Spearman ρ and top-5 overlap.

LIME: LIME explanations [12] were computed on the same 60 instances using `LimeTabularExplainer` (`num_features=10`, `num_samples=5000`). Surrogate fidelity was measured by local R^2 . The deterministic shared-instance protocol eliminates selection variance from the paired comparison.

F. Statistical Validation

Four hypothesis tests (RQ1–RQ4) were conducted (Table IV): (1) **McNemar’s test** (Yates correction) on the 82,332-sample prediction matrix; discordant pairs: $b = 1,632$ (SMOTE gains), $c = 4,784$ (SMOTE losses). (2) **Wilcoxon signed-rank** on paired confidence scores (82,332 pairs, 34,002 zero-differences excluded). (3) **Wilcoxon** on 42 paired SHAP importances. (4) **Wilcoxon** on 36 paired LIME importances.

Holm–Bonferroni correction was applied ($\alpha/4$, $\alpha/3$, $\alpha/2$, α). Effect sizes use two complementary metrics. Cohen’s h quantifies the difference between two proportions:

$$h = 2 \arcsin \sqrt{p_1} - 2 \arcsin \sqrt{p_2}, \quad (1)$$

where $|h| < 0.20$ is negligible, ≥ 0.20 small, ≥ 0.50 medium, ≥ 0.80 large. rank-biserial r is derived from the Wilcoxon statistic:

$$r = \frac{Z}{\sqrt{N}}, \quad (2)$$

where N is non-zero paired differences; $r \geq 0.10$ small, ≥ 0.30 medium, ≥ 0.50 large. Power: SHAP at $n = 42$, $r = 0.326$ gives ≈ 0.55 ; LIME at $n = 36$, $r = 0.595$ gives ≈ 0.90 . Bootstrap CIs: 95%, 2,000 resamples, percentile method.

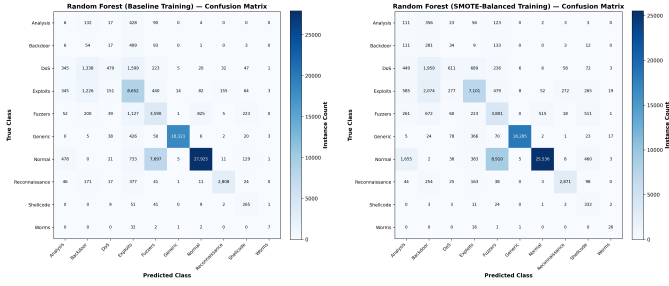


Fig. 2. Normalised confusion matrices. (left) Baseline; (right) SMOTE. SMOTE improves minority-class recall (Worms, Analysis, Backdoor) at a marginal cost to majority-class precision.

IV. EXPERIMENTAL SETUP

All experiments were implemented in Python 3.10+ using scikit-learn [21], imbalanced-learn [22], shap [10], lime [12], and SciPy [20]. The global random seed was fixed at 42. All configuration parameters are stored in `configs/*.yaml`; all output artefacts are SHA-256 verified before statistical analysis.

V. RESULTS

A. Predictive Performance (RQ1)

Table I reports per-class metrics; Table II reports 95% bootstrap confidence intervals.

SMOTE reduced accuracy by 0.0383 ($0.7543 \rightarrow 0.7161$; 95% CI Baseline $[0.7516, 0.7574]$ vs. SMOTE $[0.7129, 0.7191]$; non-overlapping) and weighted F1 by 0.0149, while improving macro F1 by 0.0170. McNemar’s test confirmed statistically significant model disagreement ($\chi^2_{\text{ Yates}}(1) = 1547.51$, $p \approx 0$; corrected $\alpha = 0.0125$). Despite statistical significance, all Cohen’s h values are negligible: $h(\text{accuracy}) = -0.0868$, $h(\text{macro F1}) = +0.0340$, $h(\text{weighted F1}) = -0.0354$ (Table III).

At the class level (Fig. 2, Table I), SMOTE improved F1 for minority attack classes (mean recall gain $+0.325$; Worms: $0.233 \rightarrow 0.452$) while degrading F1 for majority classes (mean F1 change -0.020 ; Normal: $0.848 \rightarrow 0.809$). The Worms test class contains only 44 instances; per-class F1 is presented as indicative.

Summary (RQ1): Statistically significant but practically negligible aggregate differences; minority-class F1 improved substantially while majority-class F1 declined modestly.

B. Explanation Quality (RQ2)

SHAP global rankings showed moderate stability (Spearman $\rho = 0.900$, top-5 overlap = 0.80) (Fig. 3). Wilcoxon test on 42 paired importances confirmed a statistically significant shift ($W = 283$, $p = 0.0347$; Holm $\alpha = 0.050$) with a medium effect ($r = 0.3259$) (Table IV).

LIME rankings showed considerably lower stability ($\rho = 0.565$, overlap = 0.60) (Fig. 4). Wilcoxon on 36 paired LIME importances confirmed a larger shift ($W = 115$, $p = 3.59 \times 10^{-4}$; Holm $\alpha = 0.025$) with a large effect ($r = 0.5947$) (Table IV).

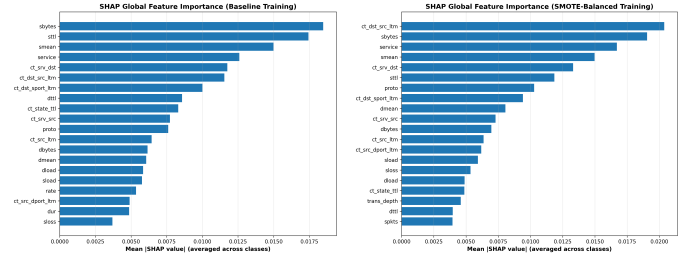


Fig. 3. SHAP global feature importance (mean $|\text{SHAP}|$, 60 instances). (left) Baseline, top: `sbytes` (0.0185). (right) SMOTE, top: `ct_dst_src_ltm` (0.0204). Spearman $\rho = 0.900$; top-5 overlap = 0.80.

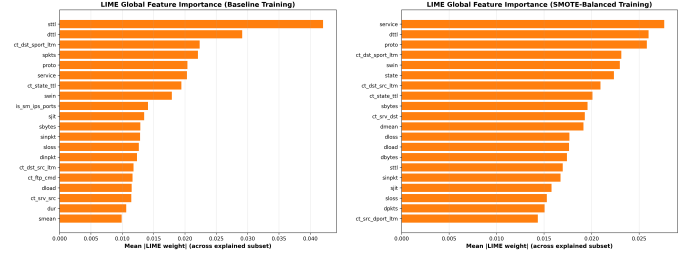


Fig. 4. LIME global feature importance (mean $|\text{weight}|$, 60 instances). (left) Baseline, top: `sttl` (0.042); $R^2 = 0.290$. (right) SMOTE, top: `service` (0.028); $R^2 = 0.252$. Spearman $\rho = 0.565$; top-5 overlap = 0.60.

LIME surrogate fidelity was 0.290 (Baseline) and 0.252 (SMOTE); the surrogate captures $\approx 25\text{--}29\%$ of local decision variance. The observed ranking shift ($r = 0.5947$) is statistically robust; absolute LIME weights should be understood as approximations. Inter-method agreement was low but improved slightly after SMOTE ($\rho = 0.410 \rightarrow 0.476$).

Summary (RQ2): SMOTE caused significant XAI explanation shifts; LIME exhibited a large effect vs. medium for SHAP.

C. Predictive vs. Explanatory Sensitivity (RQ3)

The central finding is a systematic asymmetry between predictive and explanatory sensitivity (Fig. 5, Table III). All predictive effect sizes are negligible (max $|h| = 0.0868$), while explanation effect sizes range from small (confidence scores, $r = 0.1686$) to medium (SHAP, $r = 0.3259$) to large (LIME, $r = 0.5947$). As these metrics operate on different scales and reflect different sample sizes ($n = 36\text{--}42$ feature pairs vs. $n = 82,332$ paired predictions), the comparison is qualitative: explanation attribution rankings are substantially more sensitive to class rebalancing than aggregate classification outcomes.

Summary (RQ3): Explanatory sensitivity substantially exceeds predictive sensitivity; LIME attribution rankings exhibited the greatest shift of all outputs examined.

D. Hypothesis Test Summary (RQ4)

All four tests rejected H_0 after Holm–Bonferroni correction (Table IV). The asymmetry between negligible predictive effect sizes and medium-to-large explanation effect sizes supports the research hypothesis. LIME’s greater sensitivity than SHAP ($\Delta r = 0.269$) is consistent with perturbation-based local

TABLE I
PER-CLASS CLASSIFICATION METRICS — BASELINE AND SMOTE MODELS (TEST SET, $n = 82,332$)

Class	P (B)	P (S)	R (B)	R (S)	F1 (B)	F1 (S)	n
Analysis	0.005	0.034	0.009	0.164	0.006	0.057	677
Backdoor	0.017	0.050	0.093	0.482	0.029	0.091	583
DoS	0.608	0.532	0.117	0.149	0.196	0.233	4,089
Exploits	0.625	0.788	0.777	0.638	0.693	0.705	11,132
Fuzzers	0.293	0.275	0.592	0.627	0.392	0.382	6,062
Generic	0.999	0.999	0.971	0.969	0.984	0.984	18,871
Normal	0.967	0.978	0.755	0.690	0.848	0.809	37,000
Reconnaissance	0.931	0.887	0.803	0.821	0.863	0.853	3,496
Shellcode	0.342	0.187	0.701	0.878	0.460	0.308	378
Worms [†]	0.438	0.366	0.159	0.591	0.233	0.452	44
Macro avg	0.522	0.510	0.498	0.601	0.470	0.487	82,332
Weighted avg	0.826	0.836	0.754	0.716	0.778	0.763	82,332

B = Baseline; S = SMOTE; P = Precision; R = Recall. Values rounded to 3 d.p.

[†] $n = 44$; metrics are indicative only at this sample size.

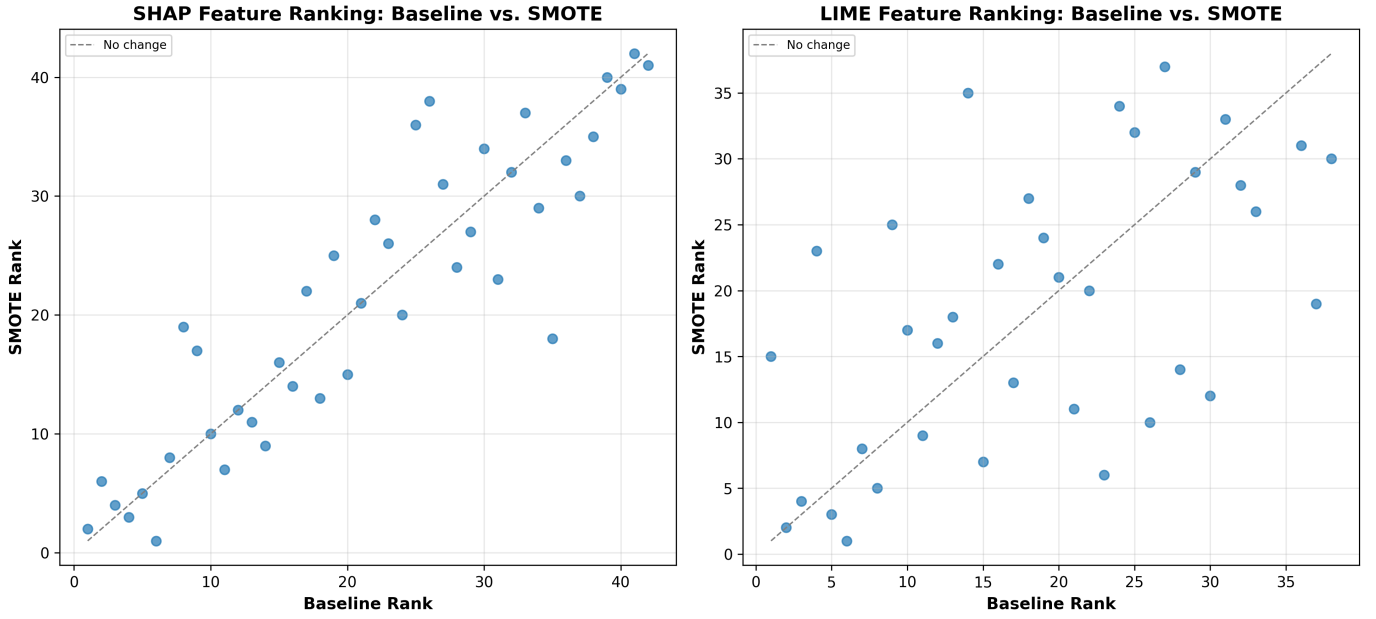


Fig. 5. Feature rank shift: Baseline vs. SMOTE for SHAP (top) and LIME (bottom). Crossing lines indicate rank changes; LIME shows substantially greater rank instability.

TABLE II
BOOTSTRAP 95% CI FOR AGGREGATE METRICS ($n_{\text{RESAMPLE}} = 2,000$, PERCENTILE)

Model	Metric	Obs.	CI _{lo}	CI _{hi}	Width
Baseline	Accuracy	0.7543	0.7516	0.7574	0.0057
Baseline	Macro F1	0.4704	0.4556	0.4853	0.0298
Baseline	Weighted F1	0.7780	0.7754	0.7808	0.0054
SMOTE	Accuracy	0.7161	0.7129	0.7191	0.0062
SMOTE	Macro F1	0.4874	0.4752	0.4989	0.0237
SMOTE	Weighted F1	0.7631	0.7605	0.7659	0.0054

Non-overlapping accuracy CIs corroborate McNemar's test.

TABLE III
EFFECT SIZES: BASELINE VS. SMOTE

Comparison	Metric	Value	Magnitude
Accuracy	h	-0.0868	Negligible
Macro F1	h	+0.0340	Negligible
Weighted F1	h	-0.0354	Negligible
Confidence shift	r	0.1686	Small
SHAP shift	r	0.3259	Medium
LIME shift	r	0.5947	Large

h : Eq. (1) ($n = 82,332$); r : Eq. (2) ($n = 42$ SHAP, $n = 36$ LIME). Thresholds (Cohen 1988): $|h| < 0.20$ neg.; $r \geq 0.10$ sm., ≥ 0.30 med., ≥ 0.50 lg. Cross-metric: qualitative only.

methods being more responsive to training distribution changes than exact tree-attribution methods.

TABLE IV
HYPOTHESIS TEST SUMMARY (HOLM–BONFERRONI, $\alpha = 0.05$)

Test	Statistic	p -value	Holm α	Reject
McNemar (Yates)	$\chi^2 = 1547.51$	$< 10^{-300}$	0.0125	Yes
Wilcoxon (conf.)	$W = 455,648,244$	$< 10^{-300}$	0.0167	Yes
Wilcoxon (LIME)	$W = 115.0$	3.59×10^{-4}	0.0250	Yes
Wilcoxon (SHAP)	$W = 283.0$	3.47×10^{-2}	0.0500	Yes

Ordered by Holm rank. $b = 1,632$, $c = 4,784$ (McNemar). SHAP power ≈ 0.55 ($n = 42$); LIME power ≈ 0.90 ($n = 36$).

VI. DISCUSSION

A. Interpretation of Findings

The central result — that SMOTE changes explanation structure more than prediction structure — follows from a mechanistic argument. Class rebalancing redistributes the model’s learned decision boundary: minority classes receive proportionally more training signal, shifting internal feature weighting even when aggregate accuracy is little changed. Post-hoc explanation methods detect this shift directly because they query the local decision surface rather than summarising it with a single metric. LIME is more sensitive than SHAP because it evaluates the model on perturbed local neighbourhoods directly affected by the shifted boundary, whereas SHAP’s global Shapley-value attributions partially average out local perturbations.

B. Practical Implications

(1) Retraining validation: Accuracy alone is insufficient. XAI outputs may change substantially even when accuracy is stable; explanation outputs must be re-validated after any retraining event. **(2) Deployment trade-off:** SMOTE gains 1,632 predictions (1.98%) but loses 4,784 (5.81%) vs. Baseline; for rare-attack-priority deployments this is acceptable, but for false-positive-sensitive environments it warrants consideration. **(3) Method selection:** A distribution change can produce a large shift ($r = 0.595$) in LIME rankings without a corresponding accuracy change; LIME-based pipelines require independent validation after retraining. **(4) Operational monitoring:** Tracking Spearman ρ of feature rankings before and after retraining detects explanation drift that accuracy-only monitoring misses.

C. Relationship to Trustworthy AI

The predictive-explanatory asymmetry has direct implications for trustworthy-AI frameworks in cybersecurity [8], [9]. Regulatory guidance increasingly requires stable explanations for consequential decisions [23]; explanations that are unstable under routine model updates become unreliable even when performance metrics remain acceptable. To the best of the author’s knowledge, this study provides the first empirical evidence of this instability in a standardised NIDS benchmark setting.

VII. LIMITATIONS

This study has the following limitations. *Internal validity:* only SMOTE was evaluated; ADASYN and cost-sensitive methods may produce different explanatory outcomes. Results are specific to Random Forest; SHAP TreeExplainer is tree-specific and LIME sensitivity may differ for neural networks. At $n = 42$, SHAP Wilcoxon power ≈ 0.55 (below the conventional 0.80 threshold); the SHAP result ($p = 0.0347$) should be interpreted with caution. At $n = 36$, LIME power ≈ 0.90 . LIME weights are linear-surrogate approximations ($R^2 \approx 0.25$ – 0.29), not exact attributions. *External validity:* UNSW-NB15 reflects 2015 laboratory conditions; transferability to modern encrypted-traffic environments warrants validation on CICIDS-2017 or CSE-CIC-IDS-2018. Concept drift in live traffic is not evaluated. *Construct validity:* explanation quality is operationalised as rank stability and magnitude shift; human interpretability and ground-truth fidelity are not assessed. SHAP and LIME rank-biserial r values are not directly commensurate.

VIII. FUTURE WORK

Future work should address: (i) alternative rebalancing techniques (ADASYN, class-weighted loss); (ii) gradient-based explanations for neural network NIDS; (iii) multi-dataset validation (CICIDS-2017, NSL-KDD); (iv) human-centred evaluation of explanation rank shifts on analyst decision-making; and (v) formal explanation-consistency metrics for model certification.

IX. CONCLUSION

A controlled empirical study on UNSW-NB15 shows that SMOTE-based class rebalancing produces statistically significant but practically negligible changes to aggregate NIDS predictive metrics ($\max |h| = 0.087$), while producing medium-to-large changes to SHAP and LIME feature attribution rankings (rank-biserial r up to 0.595 for LIME). The disproportionate sensitivity of explanation outputs relative to classification outcomes demonstrates that accuracy-only validation is insufficient for XAI-augmented NIDS pipelines.

The secondary finding — that LIME is substantially more sensitive than SHAP to training distribution changes ($\Delta r = 0.269$) — has direct implications for XAI tool selection: perturbation-based local methods must be independently validated when training data composition changes, and surrogate fidelity should be reported alongside attribution rankings.

These results support explanation-consistency monitoring as a necessary component of responsible model assessment in any NIDS pipeline that incorporates post-hoc XAI methods, particularly in regulatory contexts that require stable and reliable explanations [23].

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